

Tree functional traits as predictors of microburst-associated treefalls in tropical wet forests

Alana M. Rader¹  | Amy Cottrell² | Anna Kudla³ | Tiffany Lum⁴ | David Henderson⁵ | Harshad Karandikar⁶ | Susan G. Letcher⁷ 

¹Department of Geography, Rutgers, The State University of New Jersey, New Brunswick, New Jersey

²Forestry and Environmental Conservation Department, Clemson University, Clemson, South Carolina

³Department of Biology, Duke University, Durham, North Carolina

⁴Department of Botany, University of Hawai'i at Mānoa, Honolulu, Hawaii

⁵Division of Biology and Biomedical Sciences, Washington University in St. Louis, St. Louis, Missouri

⁶Department of Environmental Science, Policy & Management, University of California, Berkeley, Berkeley, California

⁷Department of Plant Biology, College of the Atlantic, Bar Harbor, Maine

Correspondence

Alana M. Rader, Land Change Science Research Group, Department of Geography, Rutgers University, 54 Joyce Kilmer Ave., Piscataway, NJ 08854.
Email: alana.rader@rutgers.edu

Funding information

National Science Foundation's Coastal Climate Risk and Resiliency; York International Scholars Program at Auburn University; Duke Graduate School; Middleton and Kremen Laboratories at UC Berkeley

Associate Editor: Emilio Bruna

Handling Editor: Sabrina Russo

Abstract

On 19 May 2018, a microburst caused 600 isolated forest gaps in a Costa Rican tropical forest. We surveyed fallen and standing trees within gaps to determine whether certain variables are associated with treefalls. Our results highlight considerations for future research to understand the impacts of microbursts in tropical forests.

Abstract in Spanish is available with online only

KEYWORDS

Costa Rica, functional traits, La Selva Biological Station, old growth forest, resilience, tropical storms

Thunderstorms and associated wind disturbance are becoming more common globally with a predicted 6% increase of tropical storms over the last three decades in Central America, compared to records from the 1970s (Brooks, 2013; ECLAC, 2018). In tropical

landscapes, wind disturbance forms gaps in the forest canopy, altering forest structure (Marra et al., 2014; Peterson, 2007). For example, treefall vulnerability is a product of tree damage and mortality interacting across individual and community scales. Large diameter

and tall trees are found to be more vulnerable to individual mortality directly from high-speed wind events, while small diameter and flexible subcanopy trees may be more vulnerable to indirect mortality from damaged trees around them (Everham & Brokaw, 1996; Mitchell, 2013; Woods, 2004). Additionally, vulnerability relates to the relationship between storm patterns and geographic factors such as landscape configuration and topography (Marra et al., 2014; Mitchell, 2013; Nagel & Diaci, 2006). However, the vulnerability of individual tropical trees to microbursts (localized convective events with horizontal extent of damaging winds <4 km; Hjelmfelt, 2007) has received little attention relative to larger-scale storms such as hurricanes (Chambers et al., 2007).

Treefall patterns are highly variable across tropical landscapes (Marra et al., 2018; Negrón-Juárez et al., 2018; Rifai et al., 2016), where a poorly understood source of this variability remains tree mortality mediated by microbursts. Depending on wind intensity and periodicity, tropical storms can result in landscape-altering treefalls, particularly in old growth forests, as mid- to large-stemmed, and tall trees are more prone to damage from high winds (Nagel & Diaci, 2006; Woods, 2004). Old growth trees are also impacted by lianas, seen to both increase tree vulnerability through additional weight in the forest canopy and provide resistance to falling for some individual trees through anchoring root systems and canopy stabilization (Putz, 1984). In addition to affecting vulnerability to wind damage, mechanical (i.e., root growth, tree size) and biophysical properties of forests (i.e., slope, soil properties) interact to influence damage type (Mitchell, 2013; Negrón-Juárez et al., 2010). For example, compacted soil with low infiltration limits root growth depth and increases the risk of uprooting over snapping (Negrón-Juárez et al., 2018). Alternatively, biophysical properties facilitating deeper root systems and stability may increase vulnerability of larger, less flexible trees to snapping (Mitchell, 2013).

On 19 May 2018, a large Caribbean-driven storm moved westward over La Selva Biological Station in northeastern Costa Rica, resulting in canopy gaps from a sudden microburst event (Figure 1). From 17:30 to 18:00, 38.10 mm of precipitation was recorded by the weather station at La Selva with average wind speed of 5.2 m/s against a background of 0.5 m/s. The station logged hourly average wind speed, so maximum gusts were not recorded (OTS, 2019). The microburst resulted in over 600 isolated gaps of felled old growth trees along 19.6 km of the La Selva trails and many more in the forest away from trails (O. Vargas, personal communication). Most gaps were small (c. 30 × 60 m; e.g. Figure 1b,c), consisting of one-several fallen trees (Figure 1c), characteristic of damage from localized (<4 km) outflow of convective winds (Hjelmfelt, 2007). Identifying correlates of microburst-associated mortality can improve understanding of the role that these rare, intense, small-scale storm events have on forest dynamics. We hypothesized that both physical and biological factors influence vulnerability of trees to falling during microbursts. We tested this hypothesis using four main predictions about microburst-driven treefalls: (a) trees with lianas are more likely to fall; (b) trees with buttress roots are more likely to snap than uproot; (c) tree species with higher wood densities are more likely to

uproot than snap; and (d) large trees are more likely to be part of multiple treefall events.

La Selva is located in the Caribbean lowlands of Costa Rica (10.43°N, 84.00°W), spanning approximately 1,600 hectares of tropical wet forest. We focused our study within old growth forest to control for variation in succession and disturbance vulnerability from previous land use. We surveyed approximately 4.5 km of old growth trails over 3 days, beginning 28 days after the microburst. We gathered data from all treefall events in that trail distance that met our sampling criteria, outlined below, for a total of 57 events (Table S1).

We sampled treefalls within 20 m of the trails for safe access. Although trails can affect forest dynamics, the trails we surveyed are likely to have minimal impacts as they are typically narrow (<2 m wide) and under full canopy. We sampled damaged trees ≥30 cm diameter at breast height (DBH) since trees of this size are likely to be in the canopy and contribute the majority of forest biomass (Clark & Clark, 1992). We only surveyed fallen trees that still had abundant green leaves in their canopies, to ensure that the trees studied were alive at the time of impact and to minimize the chance that these treefalls had resulted from previous disturbance.

For each treefall event, we noted whether it involved one or multiple trees. For multiple treefalls, we measured the largest tree among the fallen. Each treefall was identified as uprooted (main root system visible) or snapped (snapped trunk with root system in the ground). Because stem size is often a predictor of damage probability in strong winds (Everham & Brokaw, 1996), each fallen tree was matched to the closest standing tree within 20 m, with an equivalent DBH (±5 cm), to investigate factors affecting vulnerability. Surveyed standing trees were assumed to be subject to the same microclimate conditions as the fallen trees. DBH was measured at standard height or immediately above the buttresses (DBH mean = 65 cm) (range = 25–90). Trees were identified to the species level and assessed for liana presence and buttress size. Lianas were measured as either absent or present in the crowns of both fallen and standing trees. Buttresses were categorized ordinally from 0 to 2, where 0 represented absence of buttresses ($n = 12$), 1 indicating buttresses less than or equal to the width of the main trunk on each side ($n = 32$), and 2 indicating buttresses wider than the main trunk on each side ($n = 13$). We determined wood density of each tree species using the Tree Functional Attributes and Ecological Database (density mean = 0.63 g/m³ (range = 0.27–0.90; Harja et al., n.d.).

We evaluated how liana presence/absence impacts tree fate using Fisher's exact test, which calculates an exact p -value for a 2 × 2 table (Table S2; Zar, 2010). We evaluated whether buttress size impacts treefall using a Cochran–Mantel–Haenszel test (CMH test), appropriate for comparing ordinal versus categorical variables (Table S3; Agresti, 2012). The relationships of wood density and tree diameter to tree fate were examined using paired t tests. Two logistic generalized linear models within the subset of fallen trees analyzed (a) how the same four predictor variables may influence treefall type (uprooted or snapped) and (b) whether treefalls were single or multiple events. Continuous data were

FIGURE 1 Treefalls in old growth forests of La Selva Biological Station from multiple vantages. A comparison of Landsat 8 OLI satellite imagery, taken before (a) and after (b) a microburst event on 19 May 2018. Resulting forest damage is highlighted (b) as a decrease in near infrared reflectance, a proxy for decreased photosynthesis in vegetation (Rifai et al., 2016). The areal extent of satellite imagery within the old growth region of La Selva is highlighted in an inset map. High-resolution drone footage provides an aerial view of a single treefall event resulting from the microburst (c) (Photo credit: Dr. Paolo Segre). A photograph taken during fieldwork gives perspective of damage of a single treefall event from the ground (d) (Photo credit: Tiffany Lum)

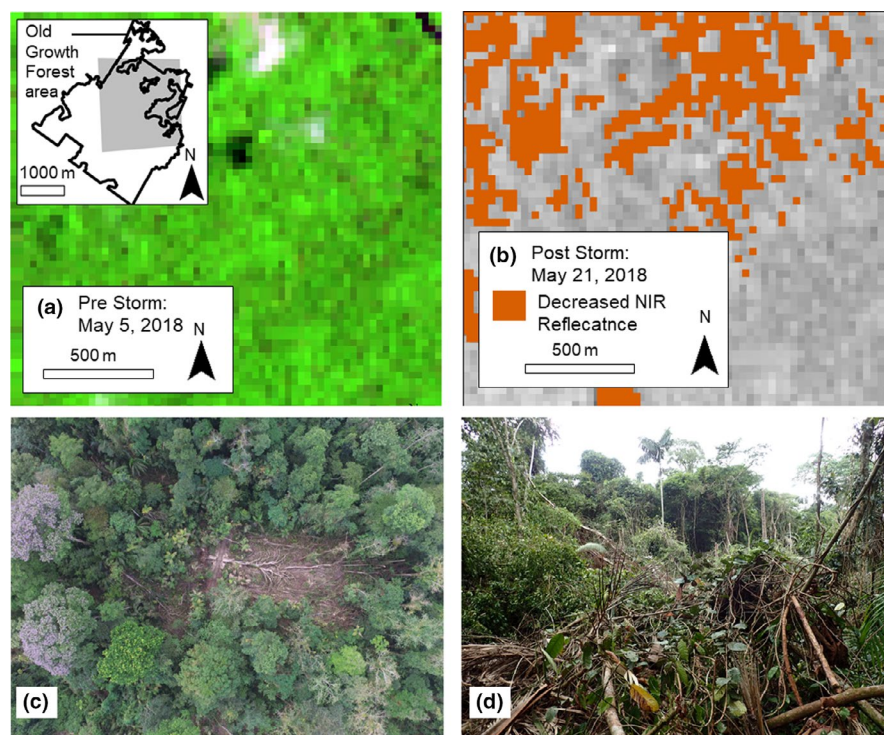


TABLE 1 Results from the logistic generalized linear models run to test hypothesis 2 (H_2) and hypothesis 3 (H_3)

	Predictors	Estimate	SE	p
H_2 : Fall Type (uproot vs. snapped)	Liana presence	0.575	±1.380	.423
	Buttress size	0.691	±0.731	.344
	Wood density	-4.419	±4.991	.376
	Tree diameter	-0.7	±0.624	.260
H_3 : Fall Magnitude (single vs. multiple)	Liana presence	1.534	±0.913	.092
	Buttress size	-0.123	±0.502	.807
	Wood density	-0.319	±3.489	.927
	Tree diameter	0.407	±0.424	.343

Note: H_2 assessed the probability of independent predictors influencing type of fall event (uproot vs. snapped). H_3 assessed the probability of independent predictor variables influencing the magnitude of a fall event (single vs. multiple). Resulting coefficient estimates and standard errors (SE) are displayed for each model.

standardized, and z-scores were calculated for wood density (g/cm^3) and tree diameter (cm) based on their mean and standard deviation. All analyses were performed in Program R version 3.5.1 (R Core Team, 2018).

Of the observed 57 treefall events, 45 involved multiple trees, and 51 had lianas present in their crown. None of the variables we quantified showed a significant relationship with fallen/non-fallen status of trees (Table S4). There was no significant difference between the fallen and non-fallen trees for liana presence (Fisher's exact test, $p = .261$), buttress size (CMH test, $X^2 = 1.79$; $df = 2$; $p = .418$), or wood density (paired t test, $t = -0.65$; $p = .522$). Within the fallen trees, none of the tested variables showed any significant relationship with treefall type, nor with single versus multiple treefalls (Table 1).

Microbursts in tropical forests are still reasonably rare events (Brooks, 2013), and the opportunity to observe their recent impacts on vegetation is uncommon. Our results indicate that at the scale and locality of our study, treefall vulnerability to microbursts and characteristics of fall events are independent of biomechanical (i.e., buttress size, wood density, diameter) and biophysical (i.e., liana presence) characteristics (Table 1). These results suggest that the impacts of microbursts may be stochastic, a finding with important ramifications for how these impacts can be incorporated into earth systems models. However, the short duration of our study precluded the detailed sampling necessary to state that conclusion with confidence. Below, we relate our findings to previous research on the factors influencing treefall dynamics across individual tree, forest community, and forest landscape scales. We also offer suggestions

for future researchers who have the opportunity to study recent microburst impacts in the field.

Although we found that characteristics of individual trees were not good predictors of tree fate in this weather event, previous research shows that individual characteristics matter. Moderate predictability of tree vulnerability is found related to size (Peterson, 2007; Ribeiro et al., 2016), wood density (Chao, Philips, Monteagudo, Torres-Lezama, & Vasques Martínez, 2009; Kraft & Ackerly, 2010), and species characteristics. For example, in a study of tree resistance to wind disturbance, Ribeiro et al. (2016) show that fall resistance increased with tree size and varies between species. Similarly, resistance of individual trees to bending increases with wood density and is controlled by species' life cycles and growth characteristics (Canham, Thompson, Zimmerman, & Uriarte, 2010; Niklas & Spatz, 2010). These studies highlight the need to expand data collection to represent a broader range of tree characteristics. Analyzing a broader range of sizes and species than we presented would allow future studies to better represent the interactions of individual characteristics in controlling tree vulnerability. Direct measurement of wood density from fallen trees is also preferable to inferring density from species identity.

Neighborhood-scale interactions matter. For example, patterns of mortality following wind disturbance in the Amazon Basin demonstrated that individual tree death both directly and indirectly affects neighboring tree mortality through increased wind exposure (Rifai et al., 2016). Assuming that the largest tree in a multiple treefall was the driver of the event, as we did, simplifies the complex relationship that trees have to each other, minimizing the impact of interacting characteristics. Further, the interactions between plant species within a forest have complex impacts on tree vulnerability to wind events. For example, Garrido-Pérez et al. (2008) relate the impact of lianas to forest age, confirming the dual roles of lianas as de-stabilizing in old growth forests and stabilizing in secondary forests. Instead of using simple presence/absence measure of lianas in the crown, future studies should attempt a more nuanced characterization of the spatial distribution of lianas. Capturing the dependence of trees on each other and other vegetation types within a forest community may provide more insight into treefall vulnerability following microbursts.

Finally, landscape matters. Treefall vulnerability relates directly to the interaction of storms and geographic patterns, such as landscape configuration, topography, and land use at the forest landscape scale (Marra et al., 2014; Mitchell, 2013; Nagel & Diaci, 2006; Rifai et al., 2016). Non-random treefall patterns across a forest landscape can result from wind flow interacting with physical geography, such as elevation and slope gradients (Espírito-Santo et al., 2010; Rifai et al., 2016). Monitoring variable meteorological conditions during storms across a range of elevations may better capture the interacting meteorological and geographic factors that impact old grown forest response to microbursts. Although sensors and monitoring are becoming cheaper and more feasible, the relative rarity of microbursts makes it unlikely that a sensor network could be deployed specifically for this purpose. For future workers studying

microburst-associated damage, we recommend taking careful notes on the slope, aspect, and landscape position of fallen trees.

This study examined whether individual tree properties influenced vulnerability of treefall during a microburst. We found that the mortality of trees ≥ 30 cm DBH was not related to the traits we measured, suggesting that treefalls during microburst are random events. However, more research is needed before that conclusion can be fully supported.

ACKNOWLEDGMENTS

The authors would like to thank the Organization of Tropical Studies and the researchers and staff of La Selva Biological Station for educational, research, and logistical support. Alana Rader acknowledges research support funding from the National Science Foundation's Coastal Climate Risk and Resiliency (C2R2) Fellowship program hosted at Rutgers University. Amy Cottrell is grateful for support funding provided by the York International Scholars Program at Auburn University. Anna Kudla appreciates The Duke Graduate School for funding through the Summer Research Fellowship. Harshad Karandikar is grateful for funding support from the Middleton and Kremen Laboratories at UC Berkeley.

CONFLICT OF INTEREST

The corresponding author confirms on behalf of all authors that there have been no conflicts of interest that might have raised bias in this research or resulting conclusions.

DATA AVAILABILITY STATEMENT

Data available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.wm37pvmhx> (Rader et al., 2019). Additionally, data will be shared and archived with the La Selva Biological Station.

ORCID

Alana M. Rader  <https://orcid.org/0000-0002-7742-2960>

Susan G. Letcher  <https://orcid.org/0000-0002-9475-7674>

REFERENCES

- Agresti, A. (2012). *Analysis of ordinal categorical data* (2nd ed.). Hoboken, NJ: John Wiley & Sons.
- Brooks, H. E. (2013). Severe thunderstorms and climate change. *Atmospheric Research*, 123, 129–138. <https://doi.org/10.1016/j.atmosres.2012.04.002>
- Canham, C. D., Thompson, J., Zimmerman, J. K., & Uriarte, M. (2010). Variation in susceptibility to hurricane damage as a function of storm intensity in Puerto Rican tree species. *Biotropica*, 42, 87–94. <https://doi.org/10.1111/j.1744-7429.2009.00545.x>
- Chambers, J. Q., Asner, G. P., Morton, D. C., Anderson, L. O., Saatchi, S. S., Espírito-Santo, F. D. B., ... Souza, C. Jr (2007). Regional ecosystem structure and function: Ecological insights from remote sensing of tropical forests. *Trends in Ecology & Evolution*, 22, 414–423. <https://doi.org/10.1016/j.tree.2007.05.001>
- Chao, K., Philips, O. L., Monteagudo, A., Torres-Lezama, A., & Vasques Martínez, R. (2009). How do trees die? Mode of death in northern Amazonia. *Journal of Vegetation Science*, 20, 260–268. <https://doi.org/10.1111/j.1654-1103.2009.05755.x>

- Clark, D. A., & Clark, D. B. (1992). Life history diversity of canopy and emergent trees in a neotropical rainforest. *Ecological Monographs*, 62, 315–344. <https://doi.org/10.2307/2937114>
- ECLAC (2018). *Climate change in Central America: Potential impacts and public policy options*. United Nations Publication. LC/MEX/L.1196/Rev.1.
- Espírito-Santo, F. D. B., Keller, M., Braswell, B., Nelson, W., Frolking, S., & Vicente, G. (2010). Storm intensity and old-growth forest disturbances in the Amazon region. *Geophysical Research Letters*, 37, L11403. <https://doi.org/10.1029/2010GL043146>
- Everham, E. M., & Brokaw, N. V. L. (1996). Forest damage and recovery from catastrophic wind. *Botanical Review*, 62, 113–185. <https://doi.org/10.1007/BF02857920>
- Garrido-Pérez, E. I., Dupuy, J. M., Durán-García, R., Ucan-May, M., Schnitzer, S. A., & Gerold, G. (2008). Effects of lianas and Hurricane Wilma on tree damage in the Yucatan Peninsula, Mexico. *Journal of Tropical Ecology*, 24, 559–562. <https://doi.org/10.1017/S0266467408005221>
- Harja, D., Rahayu, S., & Pambudi, S. (N.D.). *Wood density; Tree functional attributes and ecological database*. Retrieved from <http://db.worldagroforestry.org>
- Hjelmfelt, M. R. (2007). Microbursts and macrobursts: Windstorms and blowdowns. In E. A. Johnson, & K. Miyanishi (Eds.), *Plant disturbance ecology: The process and the response* (pp. 59–102). Burlington, MA: Academic Press.
- Kraft, N. J. B., & Ackerly, D. D. (2010). Functional trait and phylogenetic tests of community assembly across spatial scales in an Amazonian forest. *Ecological Monographs*, 80, 401–422. <https://doi.org/10.1890/09-1672.1>
- Marra, D. M., Chambers, J. Q., Higuchi, N., Trumbore, S. E., Ribeiro, G. H. P. M., dos Santos, J., ... Wirth, C. (2014). Large-scale wind disturbances promote tree diversity in a Central Amazon Forest. *PLoS ONE*, 9, e114769.
- Marra, D. M., Trumbore, S. E., Higuchi, N., Ribeiro, G. H. P. M., Negrón-Juárez, R. I., Hozwarth, F., ... Wirth, C. (2018). Windthrows control biomass patterns and functional composition of Amazon forests. *Global Change Biology*, 24, 5867–5881. <https://doi.org/10.1111/gcb.14457>
- Mitchell, S. J. (2013). Wind as a natural disturbance agent in forests: A synthesis. *Forestry*, 86, 147–157. <https://doi.org/10.1093/forestry/cps058>
- Nagel, T. A., & Diaci, J. (2006). Intermediate wind disturbance in an old-growth beech-fir forest in southwestern Slovenia. *Canadian Journal of Forest Research*, 36, 629–638.
- Negrón-Juárez, R. I., Chambers, J. Q., Guimaraes, G., Zeng, H., Raupp, C. F. M., Marra, D. M., ... Higuchi, N. (2010). Widespread Amazon forest tree mortality from a single cross-basin squall line event. *Geophysical Research Letters*, 37, L16701. <https://doi.org/10.1029/2010GL043733>
- Negrón-Juárez, R. I., Holm, J. A., Marra, D. M., Rifai, S. W., Riley, W. J., Chambers, J. Q., ... Higuchi, N. (2018). Vulnerability of Amazon forests to storm-driven tree mortality. *Environmental Research Letters*, 13, 054021. <https://doi.org/10.1088/1748-9326/aabe9f>
- Niklas, K. J., & Spatz, H.-C. (2010). Worldwide correlations of mechanical properties and green wood density. *American Journal of Botany*, 97, 1587–1594. <https://doi.org/10.3732/ajb.1000150>
- OTS (2019). Meteorological data, GPS data, and hydrological data: Organization of Tropical Studies raw data repository.
- Peterson, C. J. (2007). Consistent influence of tree diameter and species on damage in nine eastern North America tornado blowdowns. *Forest Ecology and Management*, 250, 96–108. <https://doi.org/10.1016/j.foreco.2007.03.013>
- Putz, F. E. (1984). How trees avoid and shed lianas. *Biotropica*, 16, 19–23. <https://doi.org/10.2307/2387889>
- R Core Team (2018). *R: A Language and Environment for Statistical Computing*. Vienna: R Foundation for Statistical Computing. <https://www.R-project.org>
- Rader, A. M., Cottrell, A., Kudla, A., Lum, T., Henderson, D., Karandikar, H., & Letcher, S. G. (2019). Data from: Tree functional traits as predictors of microburst-associated treefalls in tropical wet forests. *Dryad Digital Repository*, <https://doi.org/10.5061/dryad.wm37pvmhx>
- Ribeiro, G. H. P. M., Chambers, J. Q., Peterson, C. J., Trumbore, S. E., Magnabosco Marra, D., Wirth, C., ... Higuchi, N. (2016). Mechanical vulnerability and resistance to snapping and uprooting for Central Amazon tree species. *Forest Ecology and Management*, 380, 1–10. <https://doi.org/10.1016/j.foreco.2016.08.039>
- Rifai, S. W., Urquiza Muñoz, J. D., Negrón-Juárez, R. I., Ramírez ArÉvalo, F. R., Tello-Espinoza, R., Vanderwel, M. C., ... Bohlman, S. A. (2016). Landscape-scale consequences of differential tree mortality from catastrophic wind disturbance in the Amazon. *Ecological Applications*, 26, 2225–2237. <https://doi.org/10.1002/eap.1368>
- Woods, K. D. (2004). Intermediate disturbance in a late-successional hemlock – northern hardwood forest. *Journal of Ecology*, 92, 464–476. <https://doi.org/10.1111/j.0022-0477.2004.00881.x>
- Zar, J. H. (2010). *Biostatistical analysis* (5th ed.). New York, NY: Pearson.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Rader AM, Cottrell A, Kudla A, et al. Tree functional traits as predictors of microburst-associated treefalls in tropical wet forests. *Biotropica*. 2020;00:1–5. <https://doi.org/10.1111/btp.12761>